



Chapter 16

Gel-Based Assays for Measuring DNA Unwinding, Annealing, and Strand Exchange

Cody M. Rogers, Christopher W. Sausen, and Matthew L. Bochman

Abstract

Efficient replication and repair of the genome requires a multitude of protein–DNA transactions. These interactions can result in a variety of consequences for DNA such as the unwinding of double-stranded DNA (dsDNA) into single-stranded DNA (ssDNA), the annealing of complementary ssDNAs, or the exchange of ssDNA with one strand of a dsDNA duplex. Some DNA helicases possess all three activities, but many DNA-interacting proteins can also catalyze one or more of these reactions. Assays that quantify these activities are an important first step in characterizing these protein–DNA interactions *in vitro*. Here, we describe methods for the formation of dsDNA substrates and the assays that can be used to biochemically characterize proteins that can unwind, anneal, and/or exchange DNA strands.

Key words Radiolabeled DNA, DNA helicase, DNA unwinding, DNA annealing, DNA strand exchange

1 Introduction

The melting of double-stranded DNA (dsDNA) is an essential step in multiple aspects of DNA replication, recombination, and repair [1]. In the case of DNA helicases, identifying the types of substrates preferentially unwound by the protein is an informative first step in understanding how the enzyme functions. There are a plethora of techniques to measure DNA unwinding, but the complexity and output from these methods can vary considerably. Recent advances in single-molecule microscopy have allowed for analysis of DNA melting at a molecular level. Some of these modern techniques are summarized in ref. 2, and the use of magnetic tweezers to measure helicase activity is reviewed in ref. 3. However, gel-based methods for measuring DNA unwinding remain a common and highly accessible technique to assess helicase activity and can be performed with common laboratory equipment. Gel-based methods are advantageous in that they are simple to set up and flexible in the types of experiments that can be performed (e.g., time

courses vs. titrations) [4, 5]. Indeed, valuable data can be extracted from a thorough analysis of the substrates that are preferentially unwound by a helicase.

In addition to DNA unwinding, many helicases and other DNA binding proteins are capable of annealing complementary single-stranded DNA (ssDNA) strands [6]. Some helicases anneal ssDNAs so readily that DNA unwinding is masked [7]. Annealing can be observed using a method similar to measuring DNA unwinding with just a few simple modifications. Here, we describe gel-based methods to monitor DNA unwinding and annealing that we use for characterizing DNA helicases and describe helpful techniques to measure DNA unwinding by helicases that strongly anneal ssDNA [5]. Additionally, proteins involved in unwinding or annealing may also participate in DNA strand exchange [8]. Briefly, this involves swapping a strand from dsDNA with complementary ssDNA, a process which is typified by strand exchange in homologous recombination [9]. These experiments are described such that the equipment and reagents required to run each assay are nearly identical.

To begin, we describe a protocol for 5' labeling of DNA with ^{32}P and suggest alternative methods for DNA radiolabeling. Once DNA is labeled, many gel-based techniques can be used to measure protein–DNA interactions. We describe, in detail, a method for preparing reactions to observe DNA unwinding, annealing, and strand exchange. Finally, we describe how to prepare and run these gels, as well as how they can be used to quantify the desired protein–DNA transaction. Taken together, these assays are an accessible first step to characterize many protein–DNA interactions, especially in the case of DNA helicases.

2 Materials

All buffers and other reagents should be made with ultrapure MilliQ water (resistivity of 18.2 M Ω ·cm at 25 °C).

2.1 DNA Labeling

1. 10 μM oligonucleotides.
2. T4 polynucleotide kinase (T4 PNK).
3. 10 $\times$ T4 PNK buffer.
4. $\gamma[^{32}\text{P}]\text{-ATP}$ (PerkinElmer).
5. Illustra ProbeQuant G-50 micro columns (GE Healthcare).
6. EDTA: 0.5 M ethylenediaminetetraacetic acid disodium salt in water, pH to 8 with NaOH.
7. TE: 90 mM Tris–HCl (pH 8.0) and 2 mM EDTA (pH 8.0).
8. 2 $\times$ Annealing buffer: 40 mM Tris–HCl (pH 8), 8% glycerol, 0.2 mM EDTA, 80 $\mu\text{g}/\text{mL}$ BSA, 20 mM DTT, and 20 mM MgOAc.

2.2 Native Gel Electrophoresis

1. $1\times$ TBE: 90 mM Tris base, 90 mM boric acid, and 2 mM EDTA (pH 8.0).
2. 10% Ammonium persulfate (APS) in water.
3. TEMED.
4. 5 mL DNA resolving gel: 8% 19:1 acrylamide–bis-acrylamide in $1\times$ TBE, 80 μ L APS, and 20 μ L TEMED.
5. $3\times$ DNA loading dye: 15% glycerol, 60 mM EDTA, 0.15% SDS, and 0.75% bromophenol blue.
6. Polyacrylamide gel electrophoresis rig.
7. Power supply for electrophoresis.
8. 0.34 mm Whatman paper.
9. Storage phosphor screen and cassette.

2.3 Biochemical Assays

1. $2\times$ Helicase buffer: 50 mM Na-HEPES (pH 8.0), 10% glycerol, 100 mM NaOAc (pH 7.6), 300 μ M NaCl, 15 mM MgOAc, and 0.02% Tween 20.
2. 50 mM adenosine triphosphate in water, pH to 7 with NaOH (ATP).
3. 4 mg/mL SDS–Proteinase K (SDS-PK) in 50 mM Tris–HCl (pH 8.0) and 3 mM CaCl_2 .

3 Methods

3.1 DNA Substrate Preparation

1. Radiolabel 1 μ L of 10 μ M oligo 1 on the 5' end with 1 μ L T4 PNK, 2 μ L $10\times$ PNK buffer, 2 μ L $\gamma[^{32}\text{P}]\text{-ATP}$, and 14 μ L H_2O (*see Note 1*). This protocol uses a substrate with 5' and 3' ssDNA tails (i.e., fork) as an example, but a variety of substrates can be made with this method (Fig. 1a, b) (*see Note 2*). Incubate this reaction for 1 h at 37 °C. Stop the reaction with 1 μ L 0.5 M EDTA and by incubating for 5 min at 95 °C.
2. Prepare an illustra ProbeQuant G-50 micro column by spinning at $700\times g$ for 1 min to remove the storage buffer. Remove unincorporated $\gamma[^{32}\text{P}]\text{-ATP}$ by loading the entire labeling reaction onto a G-50 micro column. Place the G-50 micro column in a fresh 1.5 mL tube and spin for 2 min at $700\times g$ to collect the clean, labeled oligonucleotide, which is at ~ 450 nM assuming a 90% recovery from the G-50 micro column.
3. Set up the annealing reaction by incubating 5 μ L of 450 nM labeled oligonucleotide, 5 μ L of 500 nM complementary oligonucleotide (oligo 2), 25 μ L $2\times$ annealing buffer, and 15 μ L H_2O (*see Note 3*). Incubate the reaction at 37 °C overnight

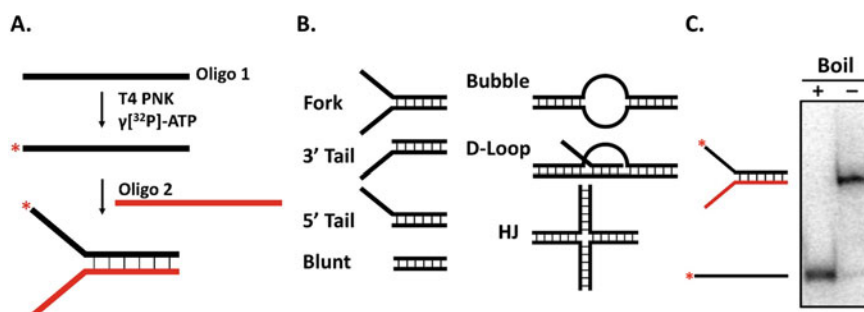


Fig. 1 Formation of DNA substrates. (a) Outline for labeling and annealing a DNA fork. Oligo 1 (black) is 5' radiolabeled (indicated by a red asterisk) prior to annealing to the partially complementary oligo 2 (red). (b) Schematic examples of the types of substrates that can be made using this protocol. (c) Representative image of labeled fork DNA made using this protocol

[10] (*see Note 4*). The desired substrate will be at ~45 nM (*see Note 5*). Dilute the annealed DNA to 10 nM in 1× TE.

4. To ensure that the DNA was properly annealed, add 10 μL of the labeled, annealed DNA to two separate tubes and add 5 μL 3× DNA loading dye to each. Incubate one of these samples at 95 °C for 5 min, which will serve as the heat-denatured reference. Load 10 μL of each dye containing sample to an 8% 19:1 acrylamide–bis-acrylamide gel. Using 1× TBE running buffer, run the gel at 10 V/cm until the dye front is near the bottom of the gel (*see Note 6*).
5. Remove the gel from the rig and place it on a piece of 0.34 mm Whatman paper. Cover the top of the gel with plastic wrap and another piece of Whatman paper. Also, place two more pieces of Whatman paper below the gel. Dry the gel on a gel dryer for 20 min at 80 °C (*see Note 7*).
6. Remove the dried gel from the sandwich leaving the plastic wrap and Whatman paper that are in contact with the gel attached. Expose the gel to a storage phosphor plate for several hours or overnight (*see Note 8*). Visualize the DNA bands using a Typhoon Imager. The heat-denatured lane (ssDNA) should migrate faster than the annealed DNA substrate. The undenatured DNA will migrate slower, but there is usually some (<5%) contaminating denatured ssDNA in the annealed sample (Fig. 1c) (*see Note 9*).

3.2 Helicase Assays

1. Set up the following reactions on ice. Dilute protein from freezer stocks to 5× the desired final concentration in 1× helicase buffer. Mix 2 μL of protein with 1 μL 10 nM labeled DNA (from Subheading 3.1), 1 μL 50 mM ATP, 1 μL H_2O , and 5 μL 2× helicase buffer (*see Note 10*). For proteins that exhibit strong annealing activity, it helps to add 10 nM cold trap, that is, a molar excess of unlabeled ssDNA that is equivalent

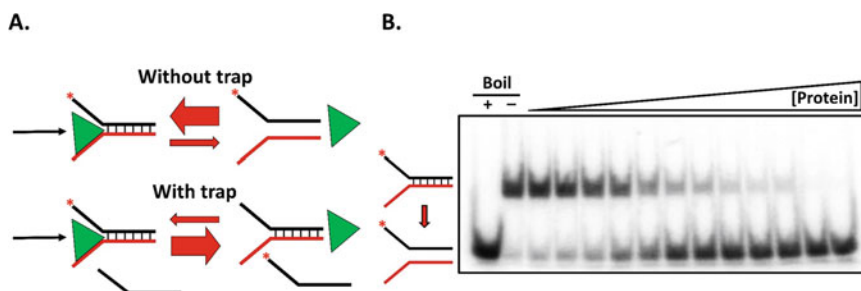


Fig. 2 DNA unwinding assay. **(a)** Complications when observing DNA unwinding by a protein with annealing activity (green triangle). In the absence of cold ssDNA trap, the unwound product can be reannealed resulting in lower than expected helicase activity. In extreme cases, the unwound product can be completely reannealed. Excess cold trap will be more likely to anneal with the unlabeled strand in the unwound product, resulting in observable labeled ssDNA. **(b)** Example of DNA unwinding. Increasing protein concentration should generate more ssDNA

to the labeled strand (oligo 1) in your substrate (Fig. 2a) (*see Note 11*). Incubate the reactions at 37 °C for 30 min (*see Note 12*). One reaction should omit protein to use as a negative control measuring spontaneously unwound DNA. Another reaction should be heat-denatured at 95 °C for 5 min to use as reference for ssDNA migration (Fig. 2b).

2. Place the unwinding reactions on ice after they are completed. To stop each reaction, add 5 μ L of 3 $\times$ DNA loading dye supplemented with 400 μ g/mL SDS-PK to each reaction. Incubate the reactions at 37 °C for 1 min to release protein from the DNA (*see Note 13*). Load 10 μ L of these reactions on an 8% 19:1 acrylamide–bis-acrylamide gel using 1 $\times$ TBE running buffer and run at 10 V/cm until the dye front reaches the bottom of the gel.
3. Dry the gel as in Subheading 3.1, step 5.
4. Image the gel as in Subheading 3.1, step 6.
5. The percent of DNA unwound can be calculated with densitometry with the following equation: (ssDNA density/total $\times$ 100). The amount of DNA unwound from the no protein control should be subtracted from each of the reactions.

3.3 Annealing Assays

1. Set up annealing reactions on ice. Dilute protein concentration from stock to 5 $\times$ the desired final concentration in 1 $\times$ helicase buffer. Prepare complementary ssDNA by heat-denaturing the desired substrate made in 3.1 for 10 min at 95 °C (*see Note 14*). Mix together 2 μ L protein, 1 μ L 10 nM melted DNA, 2 μ L H₂O, and 5 μ L 2 $\times$ helicase buffer (*see Note 15*). Incubate the reactions at 37 °C for 30 min. Omit protein from one reaction for a negative control for spontaneously annealed

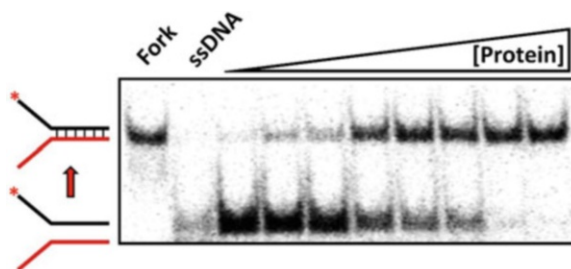


Fig. 3 Representative DNA annealing assay. This is an example of an enzyme with strong annealing activity. Increasing concentrations of proteins result in annealing of ssDNA to form the fork substrate

DNA. Another lane should contain no protein but an equimolar amount of annealed substrate as a reference for where annealed DNA migrates (Fig. 3).

2. Stop and load the reactions on a gel as in Subheading 3.2, step 2.
3. Dry the gel as in Subheading 3.1, step 5.
4. Image the gel as in Subheading 3.1, step 6.
5. The percentage of annealed DNA can be quantified using densitometry by calculating $(\text{annealed density}/\text{total} \times 100)$. The amount of spontaneously annealed DNA should be subtracted from each protein-containing reaction.

3.4 Strand Exchange Assays

1. The oligonucleotides should be designed such that the labeled, annealed DNA structure has a different migration rate than the strand exchange product (Fig. 4). These reactions should be set up on ice. Prepare protein by diluting to $5\times$ the desired final concentration in $1\times$ helicase buffer. Mix $2\ \mu\text{L}$ protein with $5\ \mu\text{L}$ $2\times$ helicase buffer, $1\ \mu\text{L}$ H_2O , $1\ \mu\text{L}$ $10\ \text{nM}$ starting substrate (e.g., fork), and $1\ \mu\text{L}$ $10\ \text{nM}$ ssDNA oligo that upon exchange makes a differentially migrating product (e.g., blunt). Incubate the reactions at $37\ ^\circ\text{C}$ for 30 min. Prepare two additional reactions that omit protein, one that contains the starting substrate and the other with the exchange product.
2. Stop and load the reactions on a gel as in Subheading 3.2, step 2 (see Note 16).
3. Dry the gel as in Subheading 3.1, step 5.
4. Image the gel as in Subheading 3.1, step 6.
5. Although there are multiple ways to quantify strand exchange, the simplest is to calculate the percent formation of the strand exchange product with the equation $(\text{exchange density}/\text{total} \times 100)$.

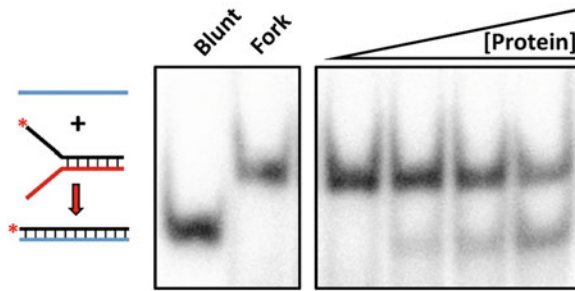


Fig. 4 Strand exchange to blunt dsDNA from a fork and a third ssDNA strand. Fork DNA (black and red) is incubated with complementary ssDNA (blue) along with increasing protein concentration. Formation of the blunt substrate (black and blue) increases in the presence of more protein

4 Notes

1. The 5' labeled DNA method described here is a simple and highly efficient technique. For those seeking other methods of radiolabeling, New England BioLabs has a nice summary of alternative techniques that use different labeling enzymes and/or other radiolabels (<https://www.neb.com/products/dna-modifying-enzymes-and-cloning-technologies/dna-labeling/dna-labeling>). While our method utilizes radioactive probes, those without access to radioactive substrates can alternatively use fluorescently labeled DNA.
2. For the simplicity of describing this method, we refer to the formation of a commonly used fork substrate. However, more complicated substrates can be made as well, including those that involve annealing more than two strands (Fig. 1b). For these substrates, incubate all of the strands together at nearly equimolar amounts. This method can also be used to make dsRNA and DNA–RNA hybrids, although the efficiency is not as high as with DNA–DNA. RNA degrades more easily than DNA, so care should be taken when using these substrates. DEPC-treated water can be used to help prevent degradation. Finally, DNA sequence is an important factor to consider when designing substrates. For example, the fork we describe has ssDNA arms that can form secondary structures that can promote or disrupt the interaction of the desired protein with DNA [11].
3. Some labs that use variations of this protocol use a greater excess of the cold oligonucleotide (oligo 2) than we show here. This can be considered in order to drive the formation of annealing to the labeled oligonucleotide. However, this means there will be some amount of contaminating, unobservable cold oligonucleotide in subsequent reactions. We have

found that using nearly equimolar DNA is sufficient for driving the formation of clean substrate.

4. We typically anneal our DNA using the overnight isothermal method we suggest here. However, another common method is to incubate the annealing reaction for 5 min in a 95 °C heat block. Then, turn the heat block off, and let it slowly cool to room temperature over several hours. This can also be done in a more controlled setting in a thermocycler.
5. The annealing reaction can be done with a higher concentration of DNA. However, a limiting factor is the efficiency of radiolabeling. Adding more DNA than what can be labeled will result in inaccurate DNA concentrations and can confound results due to the presence of excess unlabeled DNA.
6. Gel conditions should be optimized based on the structure and size of the substrate. Smaller substrates (such as blunt dsDNA) will run at nearly the same position as its unwound ssDNA counterpart on typical gels. Thus, higher percentage gels are usually preferred for these substrates. The running buffer and gel composition can be changed as well for varying effects. For example, the addition of 10 mM MgOAc to the gel can improve the stability of duplex DNA in the gel.
7. Gel percentage and thickness will affect the conditions needed to dry without cracking. We recommend running and drying a gel without substrate to determine the best parameters for drying based on the equipment available before risking the loss of a gel containing important reactions. An alternative to drying under heat and vacuum is to freeze the gels at −20 °C during the overnight exposure in a storage phosphor cassette.
8. Storage phosphor exposure duration can vary. If using a high enough concentration of DNA that was made with fresh, hot probe, several hours is usually sufficient time for good signal. However, overnight exposures are often recommended, especially when using low concentrations of DNA in biochemical assays. Longer exposures may help with older probe, but typically, the improvements to the signal are insignificant even after multiple days.
9. If the substrates are consistently highly contaminated with ssDNA (>10%), gel extraction can be used to purify the annealed substrate away from the component ssDNA. To do this, load the whole sample of prepared DNA substrate onto a gel as described in Subheading 3.1, step 4. Do not dry this gel, but instead expose the gel to autoradiography film in a dark room for 1–3 min and develop on a X-OMAT film developer. Both ssDNA and dsDNA should be observable in the same lane. Align the gel to the film image, cut out the slower migrating product, crush and place the gel fragment in 0.5× TBE,

and let diffuse overnight. Remove the buffer to a fresh tube away from the gel chunks, and perform a standard DNA precipitation. Resuspend the DNA pellet in 50 μ L TE. Continue with this substrate through Subheading 3.1, **step 4**. A substrate of known concentration will be required to determine the concentration of the gel extracted substrate, some of which will be lost during the extraction.

10. There is some flexibility in the amounts of reagents we recommend in all of these assays. For example, 5 $\times$ helicase buffer is a viable option when more than 2 μ L of protein is required to reach the desired final concentration. However, be cautious when adding protein directly from your storage buffer, especially if it contains high amounts of glycerol or is in drastically different buffer conditions than the reaction buffer. We recommend using a storage buffer similar to that used in the reaction buffer, and the helicase buffer itself can be optimized to the protein of interest. Lower (<5%) amounts of glycerol in the final reaction will improve gel images as well.
11. When using cold trap, it is helpful to set up the reactions with everything but cold trap and ATP. This will allow the protein to interact with, but not unwind, the labeled DNA of interest prior to competition by the trap. Adding the cold trap and ATP last will initiate the helicase reaction, but there are complications with method. We typically observe less overall unwinding with helicases that anneal DNA even under these conditions. Annealing is constantly in competition with unwinding, and the excess ssDNA competes for interaction with the protein. However, this method allows unwinding to be observed by helicases with exceptionally strong annealing activity [5]. For the more complex substrates with multiple points of annealing such as the bubble or Holliday junction, this method becomes even less efficient.
12. The temperature and duration of these biochemical reactions is just a recommended starting point. For uncharacterized enzymes, a range of temperatures should be tested to optimize the reaction based the protein. The method we describe is a helicase titration, which is a good place to start to determine a concentration for optimal helicase activity. However, it is often worth knowing the rate at which the helicase can unwind DNA substrates. Set up these reactions with a single concentration of protein and stop them at various time points. Most reactions are fast, so plan to take most data points from ~0 to 10 min.
13. The SDS-PK treatment we recommend here is very mild. Some enzymes may require longer treatments or higher temperatures (typically up to 50 $^{\circ}$ C). Depending on the protein, this step may be omitted entirely.

14. Alternatively, labeled ssDNA can be made as in Subheading 3.1, step 1. An equimolar amount of cold complementary oligo will need to be added to the reaction.
15. Most reports of annealing are ATP-independent, and in the case of helicases, adding ATP would inhibit annealing activity as the enzyme would unwind any annealed DNA.
16. The percentage of the gel is important here to be able to separate 3 DNA substrates: the starting substrate, the exchange product, and contaminating ssDNA.

Acknowledgments

Funding: This work was supported by the College of Arts and Sciences, Indiana University, and Indiana University Collaborative Research Grant fund of the Office of the Vice President for Research, a Collaboration in Translational Research Pilot Grant from the Indiana Clinical and Translational Sciences Institute, and the American Cancer Society [RSG-16-180-01-DMC].

References

1. Bochman ML (2014) Roles of DNA helicases in the maintenance of genome integrity. *Mol Cell Oncol* 1(3):e963429. <https://doi.org/10.4161/23723548.2014.963429>
2. Brosh RM Jr (2016) Special methods collection on DNA helicases. *Methods* 108:1–3. <https://doi.org/10.1016/j.jymeth.2016.08.009>
3. Hodeib S, Raj S, Manosas M, Zhang W, Bagchi D, Ducos B, Fiorini F, Kanaan J, Le Hir H, Allemand JF, Bensimon D, Croquette V (2017) A mechanistic study of helicases with magnetic traps. *Protein Sci* 26(7):1314–1336. <https://doi.org/10.1002/pro.3187>
4. Cejka P, Kowalczykowski SC (2010) The full-length *Saccharomyces cerevisiae* Sgs1 protein is a vigorous DNA helicase that preferentially unwinds Holliday junctions. *J Biol Chem* 285(11):8290–8301. <https://doi.org/10.1074/jbc.M109.083196>
5. Rogers CM, Wang JC, Noguchi H, Imasaki T, Takagi Y, Bochman ML (2017) Yeast Hrq1 shares structural and functional homology with the disease-linked human RecQ4 helicase. *Nucleic Acids Res* 45(9):5217–5230. <https://doi.org/10.1093/nar/gkx151>
6. Bhargava R, Onyango DO, Stark JM (2016) Regulation of single-strand annealing and its role in genome maintenance. *Trends Genet* 32(9):566–575. <https://doi.org/10.1016/j.tig.2016.06.007>
7. Macris MA, Krejci L, Bussen W, Shimamoto A, Sung P (2006) Biochemical characterization of the RECQ4 protein, mutated in Rothmund-Thomson syndrome. *DNA Repair* 5(2):172–180. <https://doi.org/10.1016/j.dnarep.2005.09.005>
8. Keller H, Kiosze K, Sachsenweger J, Haumann S, Ohlenschläger O, Nuutinen T, Syvaöja JE, Grolach M, Grosse F, Pospiech H (2014) The intrinsically disordered amino-terminal region of human RecQL4: multiple DNA-binding domains confer annealing, strand exchange and G4 DNA binding. *Nucleic Acids Res* 42(20):12614–12627. <https://doi.org/10.1093/nar/gku993>
9. Daley JM, Kwon Y, Niu H, Sung P (2013) Investigations of homologous recombination pathways and their regulation. *Yale J Biol Med* 86(4):453–461
10. Kanter DM, Kaplan DL (2011) Sld2 binds to origin single-stranded DNA and stimulates DNA annealing. *Nucleic Acids Res* 39(7):2580–2592. <https://doi.org/10.1093/nar/gkq1222>
11. Rogers CM, Bochman ML (2017) *Saccharomyces cerevisiae* Hrq1 helicase activity is affected by the sequence but not the length of single-stranded DNA. *Biochem Biophys Res Commun* 486(4):1116–1121. <https://doi.org/10.1016/j.bbrc.2017.04.003>